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Question Paper Code: 25ND014

(Regulation 2022)

Dr. N.G.P. INSTITUTE OF TECHNOLOGY

(An Autonomous Institution)

B.E./B.Tech. DEGREE END SEMESTER EXAMINATIONS, NOV/DEC - 2025

V / VII SEMESTER - B.TECH. ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

22UAD915 - ETHICS IN DATA SCIENCE

Time: 3 Hours

Maximum: 100 Marks

ANSWER ALL THE QUESTIONS

PART - A (10 x 2 marks = 20 marks)

		CO	BT Level
1	List the need of ethics for data scientists.	CO1	R
2	Define informed consent mean in data science.	CO1	R
3	Define code of ethics.	CO2	R
4	Difference between professional and corporate responsibilities of ethics in data science.	CO2	U
5	What is Algorithm Fairness?	CO3	R
6	Identify the way that discrimination occur in data analytics.	CO3	U
7	Write short note on the ethics of harms.	CO4	U
8	List the responsibilities of a professional role in data science.	CO4	R
9	How can individuals protect themselves from data harm in the internet?	CO5	U
10	List the data privacy policies.	CO5	R

PART - B (4 x 16 marks = 64 marks)

	Marks	CO	BT level
11(a) Summarize the significance of studying use cases in ethics within data science.	16	CO1	U
(OR)			
11(b) Compare ethics and law in the context of data science.	16	CO1	U
(OR)			
12(a) Analyze the ethical challenges of big data with regard to privacy. Discuss the implications for individuals and organizations and suggest strategies to address these challenges.	16	CO2	U
(OR)			
12(b) Explain the measures can be taken to ensure that autonomy is upheld in data-related practices.	16	CO2	U
(OR)			
13(a) Explain how the target marketing algorithms impact individuals and society and also identify the measures that can be taken to address these ethical concerns.	16	CO3	U
(OR)			
13(b) Compare the ethics of privacy obligations for organizations versus individuals in the context of data usage.	16	CO3	U

14(a)	Explain any one of the case-study of Harm and Bias in Data Science.	16	CO4	U
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(OR)

14(b)	Summarize the strategies to address professional role and responsibilities and harm reduction.	16	CO4	U
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PART - C (1 x 16 marks = 16 marks)

		Marks	CO	BT level
15	Evaluate the ethical issues associated with internet advertising. Discuss the role of consumer privacy, manipulation and transparency in internet advertising practices with a case study and list the strategies to address these issues.	16	CO5	An